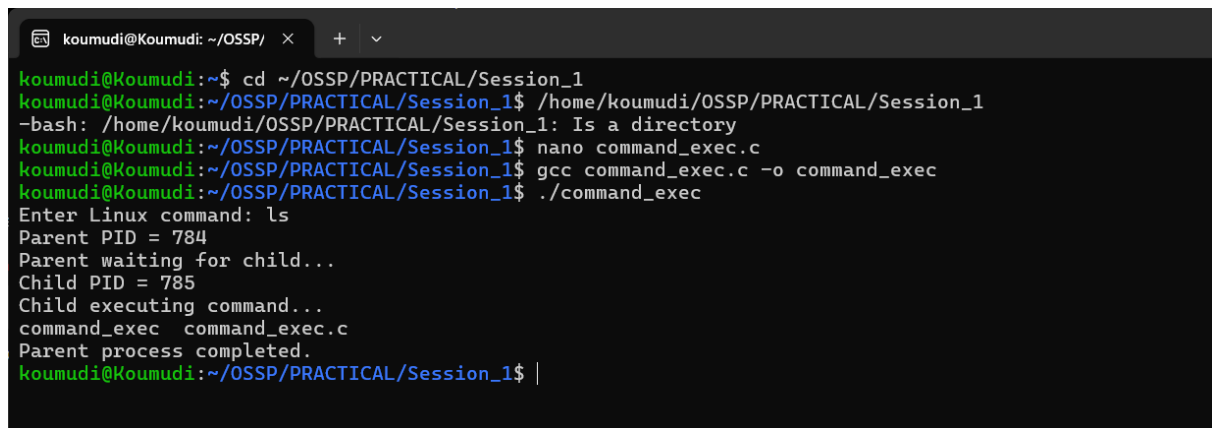


PRACTICAL (OSSP)

SESSION – 1

1. Develop a C program that:

1. Accepts a Linux command as input.
2. Creates a child process using `fork()`.
3. Executes the command in the child using an appropriate `exec()` system call.
4. Allows the parent process to wait for the child using `wait()`.
5. Displays the **Process ID (PID)** of the parent and child processes.



```
koumudi@Koumudi: ~/OSSP/ x + v
koumudi@Koumudi:~$ cd ~/OSSP/PRACTICAL/Session_1
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ /home/koumudi/OSSP/PRACTICAL/Session_1
-bash: /home/koumudi/OSSP/PRACTICAL/Session_1: Is a directory
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ nano command_exec.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ gcc command_exec.c -o command_exec
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ ./command_exec
Enter Linux command: ls
Parent PID = 784
Parent waiting for child...
Child PID = 785
Child executing command...
command_exec  command_exec.c
Parent process completed.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ |
```

2. Using Linux terminal commands: `uname` ,`lscpu`,`lsblk` ,`ps` ,`top`

Investigate the relationship between **hardware resources and operating system services** and prepare a report explaining how the OS abstracts CPU, memory, storage and I/O devices.

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ uname -a
Linux Koumudi 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64
GNU/Linux
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                  6
Model:                      186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4991.99
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
1 xsave avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
ear serialize ibt flush_l1d arch_capabilities
```

```
koumudi@Koumudi: ~/OSSP/ × + ~
Vmscape:                      Not affected
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 356.9M 1 disk
sdb  8:16   0 159.4M 1 disk
sdc  8:32   0    2G 0 disk [SWAP]
sdd  8:48   0    1T 0 disk /mnt/wslg/distro
/
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ ps
PID TTY          TIME CMD
549 pts/0        00:00:00 bash
851 pts/0        00:00:00 ps
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_1$ top
top - 16:19:22 up 24 min, 1 user, load average: 0.04, 0.02, 0.00
Tasks: 21 total, 1 running, 20 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7784.6 total,  7200.4 free,   515.0 used,   223.7 buff/cache
MiB Swap: 2048.0 total,  2048.0 free,    0.0 used.  7269.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   21876   13148   9668 S   0.0   0.2   0:01.81 systemd
    2 root        20   0    3180    2212   2072 S   0.0   0.0   0:00.04 init-systemd(Ub
    6 root        20   0    3180   20444   1940 S   0.0   0.0   0:00.00 init
   45 root        19  -1   66828   19528  18420 S   0.0   0.2   0:00.54 systemd-journal
   92 root        20   0   25020    6404   5180 S   0.0   0.1   0:01.06 systemd-udevd
  110 systemd+    20   0   21344   13236  11112 S   0.0   0.2   0:00.42 systemd-resolve
  111 systemd+    20   0   91036    7964   6996 S   0.0   0.1   0:00.23 systemd-timesyn
  171 root        20   0    4244    2688   2432 S   0.0   0.0   0:00.03 cron
  172 message+    20   0    9648    5316   4608 S   0.0   0.1   0:00.18 dbus-daemon
```